

Uncertainty in Machine Learning - WBAI054-05

Assignment 3

Total points: **80**
Starting date: May 26, 2025
Proposal Submission deadline: **23:59, May 22, 2026**
Report Submission deadline: **23:59, June 16, 2026**

Lecturer Dr. Matias Valdenegro

General guidelines:

- The tasks are targeted at groups of 2-3 students. Please make sure that the load is well divided: every student should contribute. In the case of not equal contributions that grade will be divided accordingly.
- Please take advantage of the tutorial sessions to ask your questions about the tasks.
- Provide a (short but comprehensive) explanation of what you are doing for each task.
- A reviewer should be able to understand plots independently; be sure to label axes, a legend for colors, use an easily readable font size, etc.
- Refer to all plots, tables, code blocks, etc. in your report.
- Submit the report as a single PDF file through Brightspace.
- Submit all code in a zipped file named *Code_Group_GroupNumber*, e.g., *Code_Group_20* through Brightspace.

Uncertainty Estimation Project

Maximum obtainable points: **80**

Learning Objective: The aim of this assignment is for students to build a practical implementation of a machine learning model with uncertainty estimation and perform a proper evaluation.

Step 1. Each group selects a topic for their projects and writes a short proposal (max 1 page) about what they would like to do. Consider the following information:

- General idea, Dataset(s), task, and model(s) you want to use. Set a simple goal related to the use of uncertainty estimation.
- Short description of tasks you will do (train models, use XYZ evaluation metrics, etc)

Some general project ideas are available below:

- Train a model over a dataset with proper uncertainty estimation and provide a correct evaluation.
- Perform out of distribution detection using two datasets and a single model.
- Validate some properties of uncertainty like: epistemic uncertainty changing with training set size, aleatoric uncertainty constant with training set size, uncertainty varying with class imbalance, degradation of calibration with corrupted features (like Ovadia et al 2019).

You can also propose ideas of your own that are outside what is said above, but close enough, you can ask the coordinator via email.

Your project must include programming and you should train machine learning models using python ML/DL framework of your choice (tensorflow, keras, pytorch, etc). The project should involve training or fine-tuning a model with uncertainty estimation.

Consider that the project should take around 20-25 hours of your time. Submit the proposal as a PDF file in Brightspace. You will get feedback and accept/reject within a couple of days. If you have doubts about your project being accepted before submitting, you can ask Matias via email.

Rejection might come because the project is too simple or difficult or might take you too much more than the allocated time, and then changes can be made to adapt requirements.

Step 2. Once your project topic is accepted, start working on your project in your time and during the practicals. Attend practicals to ask questions to TAs or via email to the coordinator.

Step 3. By the deadline the project must end and you should submit a written report (3-10 pages), indicating what you did in your project, results, analysis, to what degree you fulfilled your goal, and include source code as attachments. Any source code you reused from past projects or from the internet can be used but must be properly cited and referenced. It should be clear what is new for this project and what is reused from the internet.

Make sure that your report is readable and understandable, and that it contains all necessary details for someone else to reproduce your results. Any use of material should be clearly cited and quoted/referenced, including any paper(s) that you might use as basis.

Evaluation Rubric

What we will evaluate and in what weight is described below:

- 30% Project quality and consistency with the learning outcome and use of uncertainty estimation. Did the group train a model on a task of decent difficulty? (not a toy dataset like MNIST) Did the group succeed to a certain degree with respect to their own goals? Failure is not unexpected and generally acceptable, if students tried enough (model variations, different datasets, preprocessing, etc) this also counts as success. Failure to achieve the task without explanation or further reasons will decrease the grade significantly. The project should be approx 20-25 hours of work. Significantly less than this standard will decrease the grade.
- 30% Model training. Are the models correctly trained? Is the task setup correctly? Are uncertainty methods correctly implemented? Does the evaluation make sense and is appropriate for the task?.
- 40% Report structure and content, citations, figure/table referencing, proper reuse of previous work. Overall grade for the format, structure, and content of the report, does it follow like a scientific paper? Are the steps performed by students well described? Is the project reproducible to a certain degree? (We cannot expect full reproducibility, but only to a good amount). There is no need to describe the general methods that were used if they are covered in the lectures, only what is specific to the project.

The report should be understandable and clearly written/structured, we will not be checking grammar or typos, but overall description of the project. If parts are not understandable, points can be deducted or marked as incorrect.